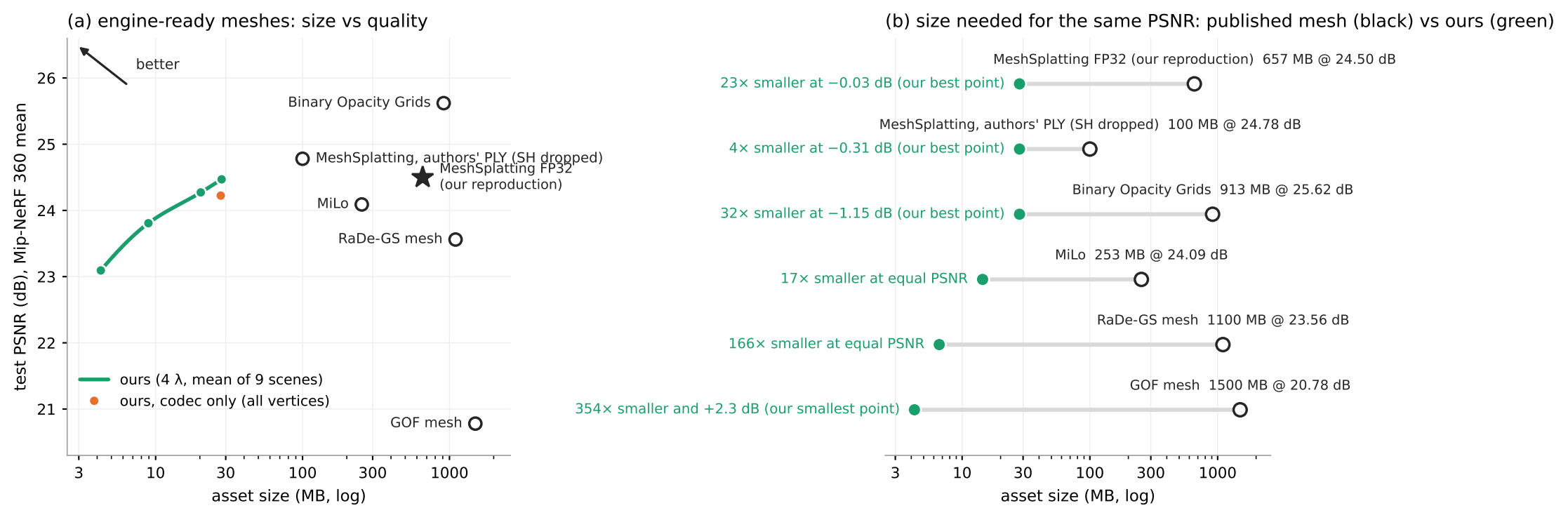
(a) engine-ready meshes: size vs quality
better
Binary Opacity Grids
MeshSplatting, authors' PLY (SH dropped)
MeshSplatting FP32
(our reproduction)
MiLo
RaDe-GS mesh
GOF mesh
ours (4 λ, mean of 9 scenes)
ours, codec only (all vertices)
test PSNR (dB), Mip-NeRF 360 mean
asset size (MB, log)
(b) size needed for the same PSNR: published mesh (black) vs ours (green)
MeshSplatting FP32 (our reproduction) 657 MB @ 24.50 dB
23× smaller at −0.03 dB (our best point)
MeshSplatting, authors' PLY (SH dropped) 100 MB @ 24.78 dB
4× smaller at −0.31 dB (our best point)
Binary Opacity Grids 913 MB @ 25.62 dB
32× smaller at −1.15 dB (our best point)
MiLo 253 MB @ 24.09 dB
17× smaller at equal PSNR
RaDe-GS mesh 1100 MB @ 23.56 dB
166× smaller at equal PSNR
GOF mesh 1500 MB @ 20.78 dB
354× smaller and +2.3 dB (our smallest point)
asset size (MB, log)